Lab 1: Musical Genre Classification

Machine Learning for Sound and Music

Lucía González Pérez

Universitat Pompeu Fabra
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1. Introduction

The aim of this lab is to evaluate different machine learning algorithms for music genre classification using datasets with different audio segment durations. The experiments compare Logistic Regression, Artificial Neural Networks, k-Nearest Neighbors (k-NN), and Decision Trees, analysing their performance through training accuracy, cross-validation accuracy, overfitting, and confusion matrices. We will discuss whether short audio segments contain enough information to classify music genres reliably.

2. Code implementation

The code has been developed in a python environment using Jupyter notebook. There are three main files:

- model_definitions.py defines the machine learning pipelines and hyperparameter grids for Logistic Regression, k-NN, Decision Trees, and Artificial Neural Networks. Each dataset has its own configurations, including preprocessing, feature selection, and model parameters.
- run_lab1.py implements the main experimental workflow. It loads and preprocesses the datasets, performs hyperparameter searches using GridSearchCV, calculates training and cross-validation accuracy and the overfitting gap, and generates confusion matrices using out-of-fold predictions. It also allows the user to export results to Excel and save confusion matrix figures as PNG files.
- run_lab1.ipynb allows the user to select a dataset and one or more algorithms, evaluate their configurations using stratified 10-fold cross-validation, display the results, and generate a confusion matrix for a selected configuration. Results can be saved as .xlsx and .png. The required CSV datasets must be available in the Lab1/data directory.

URL: https://github.com/lgonperez3-glitch/MLSM-labs/tree/main/Lab1

3. Results

Table 1 shows the best configuration for each dataset, together with its parameters and performance metrics. More results are shown in Lab1/results directory. The confusion matrix for each of the results in table 1 are the shown in figures 1, 2 and 3

Table 1. Best trained algorithm for each dataset

Dataset	Algorithm	Parameters	Train accuracy	CV accuracy	Overfitting gap
GenreAll	Logistic Regression	{'model__C': 0.2, 'selector__k': 150}	0,9803	0,8590	0,1213
30 sec	Artificial Neural Network	{'model__alpha': 0.001, 'model__hidden_layer_sizes': (128, 64), 'model__learning_rate_init': 0.001, 'selector__k': 'all'}	0,9991	0,7620	0,2371
3 sec	k-NN	{'model__n_neighbors': 5, 'model__p': 1, 'model__weights': 'distance', 'selector__k': 'all'}	0,9991	0,9267	0,0724

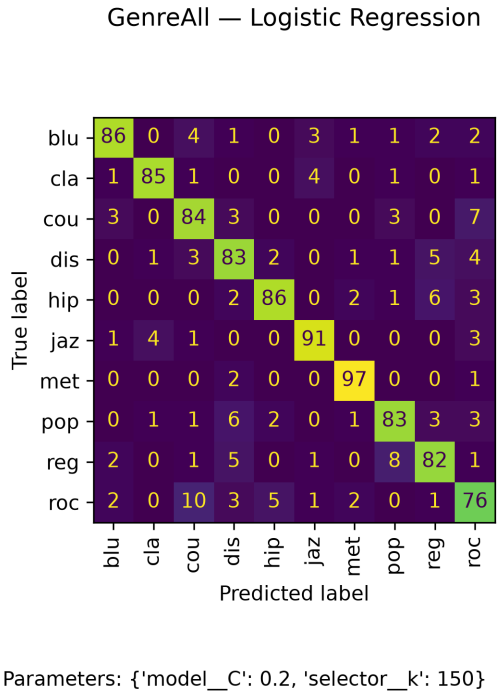
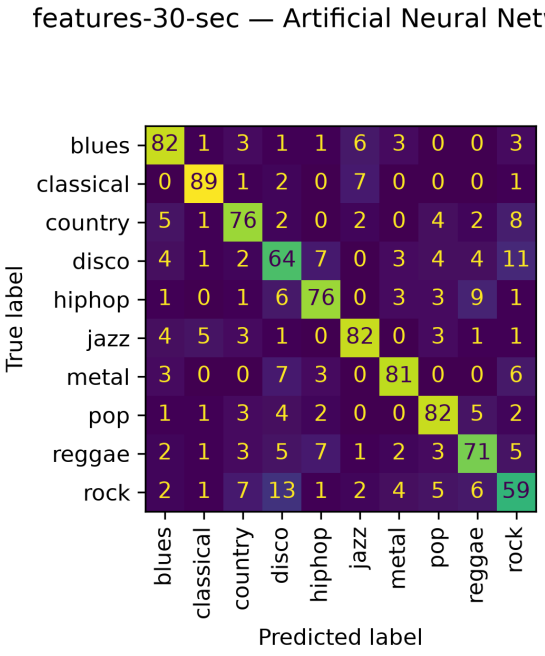
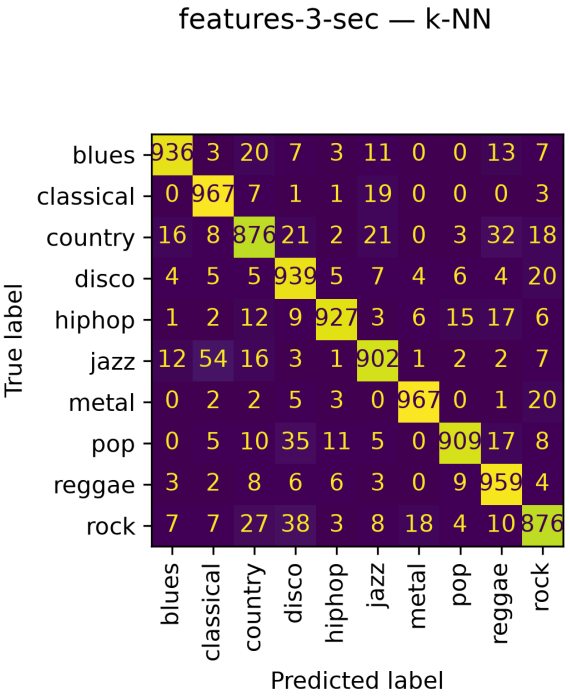


Fig 1. GenreAll - Logistic regression



Parameters: {'model__alpha': 0.001, 'model__hidden_layer_sizes': (128, 64), 'model__learning_rate_init': 0.001, 'selector__k': 'all'}

Fig 2. Features 30 seconds - ANN



Parameters: {'model__n_neighbors': 5, 'model__p': 1, 'model__weights': 'distance', 'selector__k': 'all'}

Fig 3. Features 3 seconds - k-NN

4. Discussion

Is the 3 s accuracy real or inflated?

The accuracy with 3-second segments is probably higher than it would be when testing songs that the model has never seen before. This is because segments from the same song can end up in both the training and validation sets. This could explain why k-NN gets such a high accuracy.. The model might be recognising similarities between segments of the same song, instead of learning patterns that work well with completely new songs. So, the accuracy is correct for the way we tested the model, but it might not show how well it works with new songs.

Are 3 seconds enough to identify a genre?

The results suggest that 3 seconds can contain quite a lot of information about the music genre. k-NN and the neural network get more than 80% of accuracy. However, this does not necessarily mean that 3 seconds are enough to classify new songs correctly. Since segments from the same song can appear in both the training and validation sets, the results might be better than they would be with completely new songs. So, 3-second segments seem useful, but we cannot be sure how well they work on songs the models have never seen.

Are the differences between models significant?

- 3-second dataset: The k-NN model gets the best accuracy at 92,67%, followed by the neural network at 86.97%. Logistic regression gets 72.50%, while the decision tree gets 65.91%.
- 30-second dataset: The neural network gets the best accuracy at 75.20%, followed by logistic regression at 73.70%, k-NN at 72.50%, and the decision tree at 59.50%.The differences between the neural network, logistic regression, and k-NN are relatively small.
- GenreAll dataset: The artificial logistic regression gets the best accuracy at 85.90%, followed closely by the artificial neural network at 81.47%. k-NN gets 78.33%, while the decision tree gets 65.35%.The neural network and logistic regression perform very similarly, so it is difficult to say which one is truly better based only on these results. k-NN performs worse, and the decision tree gets the lowest accuracy.

Which genres are most similar given these descriptors?

From figures 1, 2 and 3, we can see that the most similar genres are:

- Blues - Country - Jazz
- Classical - Jazz
- Country - Rock - Blues
- Disco - Pop - Rock

This conclusion is made since confusion matrices show the examples from the dataset that have been misclassified, meaning that these examples could have similarities to other genres.

5. Conclusion

The experiments show that the best-performing algorithm depends on the dataset and audio segment duration. k-NN achieves the highest reported cross-validation accuracy on the 3-second dataset, while the Artificial Neural Network performs best on the 30-second dataset. Logistic Regression obtains the best result on GenreAll. However, the high accuracy observed with short segments should be interpreted cautiously, as segments from the same song may appear in both training and validation folds. Further evaluation using song-level separation would be necessary to determine how well the models generalise to previously unseen songs.